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Table of content

Solution.....**3**

AI Solution.....**4**

Business objectives.....**5**

Problem definition.....**11**

Machine learning approach.....**12**

Data.....**14**

Model.....**17**

Time series analysis on data.....**19**

Solution.....**21**

Natural Language Processing.....**23**

Deep learning.....**26**

Other features.....**27**

Poster.....**29**

Declaration and references.....**30**

Grammly report.....**31**

Semifori Elec Tech (Smart Light AI)

Semifori Elec Tech (Smart Light AI) is an artificial intelligence-based streetlight monitoring and fault-detection system aligned with the Fourth Industrial Revolution (4IR) through machine learning and data analysis. It addresses challenges faced by South African municipalities, including delayed fault detection, high maintenance costs, electricity waste, and poor streetlight maintenance.

Many municipalities rely on residents to report faulty streetlights or on maintenance teams to conduct physical inspections. This results in delayed repairs, increased costs, and reduced public safety, particularly at night.

Smart Light AI provides an intelligent solution that analyses streetlight sensor data, including voltage, power consumption, current fluctuations, and temperature, to automatically detect and classify faults such as bulb failures, short circuits, voltage surges, and flickering.

The system uses a machine learning model to identify faults and displays its predictions on a monitoring dashboard. This helps municipalities identify problematic streetlights, prioritize maintenance, reduce electricity waste, and improve infrastructure reliability.

The main objective is to support smarter, more efficient streetlight maintenance while promoting safer, more sustainable communities.

AI SOLUTION

Semifori Elec Tech (Smart Light AI)

Our proposed AI solution is an intelligent streetlight and electricity fault detection system that uses machine learning to identify and classify electrical faults. The system analyses streetlight operating data, including voltage, current fluctuations, power consumption, and temperature, to detect abnormal conditions and potential failures. Using Python and a Random Forest machine learning model, the system identifies faults such as bulb failures, short circuits, voltage surges, and flickering. The results are displayed on a monitoring dashboard, helping municipalities identify problematic streetlights and prioritise maintenance. This solution is relevant to the Fourth Industrial Revolution (4IR) because it applies artificial intelligence, automation and data analysis to improve traditional municipal operations. It also supports environmental sustainability by helping to reduce electricity waste, improve energy efficiency, and minimise unnecessary maintenance costs. Ultimately, Smart Light AI aims to help municipalities detect infrastructure problems earlier, improve maintenance responses and create safer, more sustainable communities.

Business objectives:

- Business Background

Local municipalities in South Africa face significant challenges in maintaining public streetlight infrastructure and electrical distribution networks. Faults such as burnt-out bulbs, voltage drops, short circuits, cable disruptions, and equipment overheating can compromise public safety, increase maintenance costs, and contribute to electricity wastage. Traditional maintenance processes often depend on residents reporting faulty streetlights or maintenance teams conducting physical inspections. This can delay fault identification and repairs.

Semifori Elec Tech (Smart Light AI) provides an intelligent streetlight monitoring and fault detection solution that uses Artificial Intelligence (AI), machine learning, and IoT technology to improve municipal infrastructure management. The system analyses streetlight operating data, including power consumption, voltage levels, current fluctuations, temperature and environmental conditions, to identify abnormal patterns and classify potential electrical faults.

Using a Random Forest machine learning model developed in Python, the system identifies different streetlight fault conditions and presents the results through an interactive Streamlit monitoring dashboard. This enables municipalities to monitor infrastructure performance, identify problematic streetlights and prioritise maintenance activities. The solution promotes proactive maintenance, improved energy efficiency and safer communities.

- Business Objectives

The main business objectives of Semifori Elec Tech (Smart Light AI) are:

- **Automate fault detection:** Use artificial intelligence and machine learning to analyse streetlight operating data and automatically identify abnormal conditions and potential electrical faults.
- **Improve fault identification** by classifying streetlight conditions into five categories: No Fault, Bulb Failure, Short Circuit, Voltage Surge, and Light Flickering.

- **Reduce maintenance delays:** Help municipalities identify faulty streetlights earlier and prioritise maintenance activities to improve repair response times.
- **Improve infrastructure monitoring:** Provide a centralised monitoring dashboard that displays streetlight records, AI predictions, and maintenance statuses.
- **Reduce electricity wastage:** Identify streetlight faults and abnormal operating conditions that may contribute to unnecessary electricity consumption.
- **Optimise maintenance resources:** Help municipalities reduce unnecessary physical inspections and allocate maintenance teams more efficiently.
- **Support data-driven decision-making:** Provide fault classification results and infrastructure information that assist municipal supervisors with maintenance planning and resource allocation.
- **Promote predictive maintenance:** Use historical streetlight operating data and identified patterns to support the early identification of potential equipment failures.

- Business Success Criteria

The Semifori Elec Tech system will be considered successful when the following criteria are achieved:

- Successful collection, loading, and processing of streetlight operating data.
- Accurate preprocessing of sensor information for machine learning analysis.
- Successful training and execution of the Random Forest classification model.
- Ability to classify streetlight conditions into all five supported categories.
- Evaluation of model performance using accuracy, precision, recall, and a confusion matrix.
- Reliable identification and display of predicted streetlight faults.

- Successful operation of the Streamlit dashboard, including streetlight records, AI predictions, and maintenance statuses.
- Effective identification of faulty streetlights to support municipal maintenance planning.
- Successful execution of the complete system workflow:
- Streetlight Data Collection → Python Data Processing → Machine Learning Analysis → Fault Classification → Monitoring Dashboard → Maintenance Planning.

-Functional Requirements

The Semifori Elec Tech system must satisfy the following functional requirements:

- **FR-01 (Sensor Data Collection):** The system must collect streetlight operating information, including voltage, current, power consumption, temperature, and environmental conditions.
- **FR-02 (Data Preprocessing):** Python scripts must clean, organise and prepare streetlight data for machine learning analysis.
- **FR-03 (Feature Processing):** The system must process relevant sensor features and convert environmental conditions into numerical values where required.
- **FR-04 (Machine Learning):** The system must use a trained Random Forest classification model to analyse streetlight operating conditions.
- **FR-05 (Fault Classification):** The machine learning model must classify streetlight conditions into No Fault, Bulb Failure, Short Circuit, Voltage Surge and Light Flickering.
- **FR-06 (Prediction Display):** The system must display AI-generated predictions alongside the corresponding streetlight records.
- **FR-07 (Dashboard Monitoring):** The Streamlit dashboard must allow users to view streetlight records, predicted faults, and monitoring results.
- **FR-08 (Fault Filtering):** The dashboard must allow users to filter records to focus on streetlights predicted to have faults.

- **FR-09 (Maintenance Status):** The system must display maintenance-related statuses based on predicted streetlight conditions.
- **FR-10 (Model Evaluation):** The system must provide model evaluation results to assess classification performance and identify areas requiring improvement.
- **FR-11 (Fault Reporting):** The system should support the generation of structured fault reports containing relevant streetlight information and predicted fault categories.
- **FR-12 (Municipal Notifications):** The system should support future integration with municipal maintenance platforms to communicate detected faults and assist maintenance teams.

-Non-Functional Requirements

The Semifori Elec Tech system must satisfy the following non-functional requirements:

- **Accuracy:** The machine learning model should achieve reliable classification results while minimising incorrect predictions and false alarms.
- **Reliability:** The system must process available streetlight data consistently and maintain stable operation.
- **Performance:** Data processing, machine learning predictions, and dashboard operations should execute within a reasonable time.
- **Usability:** The dashboard should have a simple and intuitive interface accessible to both technical and non-technical municipal employees.
- **Maintainability:** Python scripts and system components should be organised to support debugging, updates and future improvements.
- **Security:** Streetlight datasets and model files should be protected against unauthorised access and modification.
- **Scalability:** The system should support additional streetlight records, fault categories and future integration with larger municipal infrastructure networks.

- Constraints

The development and implementation of Semifori Elec Tech will be subject to the following constraints:

- **Hardware Availability:** Access to physical IoT sensors and installation equipment may affect the deployment of live streetlight monitoring.
- **Data Availability:** The effectiveness of machine learning predictions depends on the quality, quantity, and distribution of available streetlight operating data.
- **Budgetary Limits:** Hardware, infrastructure and deployment costs may influence the expansion of the system.
- **System Integration:** Connecting the system to existing municipal maintenance management platforms may require additional technical resources and permissions.
- **Data Security:** Streetlight operating information must be protected to maintain data integrity and prevent unauthorised access.
- **Infrastructure Compatibility:** Integration with different municipal streetlight systems may require adjustments to data formats and communication technologies.

- Risk Matrix

Identified Risk	Potential Impact	Mitigation Strategy
Corrupted or incomplete sensor data	Incorrect predictions and processing errors	Apply data validation, cleaning and preprocessing techniques.
Class imbalance	Poor detection of less common fault categories	Analyse class distribution and apply appropriate data-balancing techniques.
High false-positive rate	Healthy streetlights may be incorrectly classified as faulty	Evaluate precision, recall and confusion matrix results and improve model performance.
Incorrect fault classification	Maintenance decisions may be based on inaccurate predictions	Test the model using unseen data and evaluate individual fault categories.
Sensor communication failure	Missing or delayed streetlight operating information	Implement communication monitoring and error-handling mechanisms.
Model performance degradation	Reduced prediction reliability as operating conditions changes	Regularly evaluate and retrain the model using updated data.
Unauthorised data access	Compromised infrastructure information	Implement appropriate access controls and data protection measures.
Integration difficulties	Delays in connecting with municipal infrastructure systems	Use standardised data formats and conduct integration testing.

Problem definition:

What is the problem?

Many communities in South Africa experience faulty streetlights and electricity-related infrastructure problems that are not detected or reported immediately. Often, faults are only identified when residents notice malfunctioning streetlights and report them to their local municipality. This results in delays in fault detection and maintenance, potentially compromising public safety, increasing maintenance costs, and contributing to unnecessary electricity wastage.

How relevant is it to the theme?

The problem is directly relevant to the Fourth Industrial Revolution (4IR) because it presents an opportunity to apply Artificial Intelligence (AI), the Internet of Things (IoT), machine learning and automation to improve traditional municipal services. IoT sensors can collect streetlight operating information, including voltage, current, temperature and operating status. Python can process and analyse this information, while machine learning can identify abnormal patterns and classify potential faults. Semifori Elec Tech applies these technologies to modernise streetlight monitoring and fault detection.

How beneficial will it be in solving the problem?

Semifori Elec Tech will benefit municipalities and communities by improving fault detection, supporting faster maintenance responses and reducing dependence on manual reporting. The system will help municipalities identify problematic streetlights, prioritise maintenance activities, minimise unnecessary electricity consumption and use maintenance resources more efficiently. Ultimately, it aims to improve public safety, strengthen municipal service delivery and promote environmental sustainability.

Section 2

AI SOLUTION – THEORETICAL ASPECT

Smart Light AI: Intelligent Streetlight Fault Detection and Maintenance Support System

1. Machine Learning Approach

Smart Light AI uses supervised machine learning to identify and classify streetlight faults using electrical and environmental sensor readings. The approach is relevant to the proposed solution because streetlight faults can be associated with changes in power consumption, voltage, current, and temperature.

The system uses historical labelled sensor data to train classification models. These models learn relationships between sensor measurements and known streetlight fault categories. Once trained, they can analyse new readings and predict the possible condition of a streetlight.

The project implements and compares several machine learning algorithms to identify a suitable model for the fault detection problem.

The algorithms include:

- **Logistic Regression:** Used as a classification algorithm and baseline for comparing model performance.
- **Random Forest:** Uses multiple decision trees to identify patterns in sensor readings and classify streetlight faults.
- **Support Vector Machine (SVM):** Evaluates how effectively different fault categories can be separated using sensor measurements.
- **Gradient Boosting:** Combines sequential predictive models to improve classification performance.
- **Multi-Layer Perceptron (MLP):** Introduces a neural network approach for learning complex relationships between sensor measurements and fault categories.
- **Majority Classifier:** Provides a baseline against which more advanced models can be evaluated.
- **Augmented Random Forest:** Investigates whether improving the representation of underrepresented fault categories can improve prediction performance.

Random Forest is used as the main classification approach in the monitoring application, while the other algorithms provide a basis for model comparison.

This approach is appropriate because the system must distinguish between several possible streetlight conditions rather than simply identifying whether a streetlight is working or not.

2. Data

Smart Light AI uses a structured streetlight fault prediction dataset containing 34,310 records and nine columns. The dataset provides the information required to train, evaluate, and demonstrate the machine learning solution.

The data consists of electrical measurements, environmental information, timestamps, streetlight identification numbers, and corresponding fault categories.

https://www.kaggle.com/code/vizeno/street-light-fault-prediction

2.1 Data fields

Data field	Description	Relevance to the solution
bulb number	Streetlight identification number	Identifies individual streetlights
timestamp	Date and time of a reading	Supports monitoring and time-series analysis
power consumption (Watts)	Electricity consumption	Helps identify unusual power usage
voltage levels (Volts)	Voltage measurement	Helps identify voltage-related problems
Current fluctuations (Amperes)	Electrical current variation	Helps identify abnormal electrical behaviour
temperature (Celsius)	Temperature measurement	Provides information about possible overheating
environmental conditions	Environmental conditions	Provides context for sensor readings
Current fluctuations env (Amperes)	Environment-adjusted current fluctuations	Provides an additional measurement for classification

Fault type	Recorded streetlight condition	Provides the target category for supervised learning

2.2 Types of data used

The system uses different forms of information that contribute to streetlight fault detection:

Numerical data: Power consumption, voltage, current fluctuations, and temperature are numerical measurements used as machine learning input features.

Categorical data: Environmental conditions and fault types represent categories that can be processed by the system.

Time-based data: Timestamps provide chronological information that supports the analysis of sensor readings and fault patterns over time.

Identification data: Streetlight asset numbers allow the system to distinguish between individual streetlights and associate monitoring results with assets.

2.3 Fault categories

The dataset contains five classification categories:

1. No Fault.
2. Short Circuit.
3. Voltage Surge.
4. Bulb Failure.
5. Light Flickering.

These categories allow the machine learning model to identify different streetlight conditions and provide more useful information than a simple working/not-working classification.

2.4 Data preparation

The dataset is prepared before it is used by the machine learning algorithms. Environmental conditions are converted into numerical representations where required. The selected sensor measurements are used as input features, while the fault category is used as the target variable.

The project also considers class distribution and includes data augmentation experiments to investigate ways of improving the representation of less frequent fault categories.

The dataset therefore provides the foundation for training the models and demonstrating how sensor information can be used to identify possible streetlight faults.

3. Model

The developed AI model must be evaluated to determine how accurately it identifies streetlight faults. Smart Light AI uses multiple evaluation metrics to assess model performance and understand how well it classifies the different fault categories.

Accuracy is included, but the project does not depend on accuracy alone. This is important because some fault categories may have fewer records than others.

The following evaluation techniques are implemented:

Accuracy: Measures the percentage of predictions that match the actual fault categories.

Balanced Accuracy: Evaluates classification performance across the different categories while accounting for class imbalance.

Precision: Measures how many predictions made for a particular fault category are correct.

Recall: Measures how many actual examples of a fault category are successfully identified by the model.

Macro F1-score: Measures the average F1-score across all fault categories, giving each category equal importance.

Weighted F1-score: Calculates the F1-score while considering the number of records belonging to each category.

Matthews Correlation Coefficient (MCC): Provides an additional measure of the quality of classification predictions.

Confusion Matrix: Compares actual fault categories with predicted categories and identifies where the model incorrectly classifies readings.

ROC-AUC: Provides an additional measure of how effectively the model distinguishes between categories where the calculation is supported.

The project also uses Stratified K-Fold cross-validation to assess model performance across multiple data splits.

These evaluation methods are used to compare candidate algorithms and determine whether the selected model is suitable for the streetlight fault detection problem.

The Model section of the dashboard presents available evaluation results, allowing users to examine model performance and understand its strengths and weaknesses.

4. Time Series Analysis on Data

Time-series analysis is an important part of Smart Light AI because streetlight sensor measurements can change over time. Examining individual sensor readings provides information about a streetlight at a particular moment, while analysing readings over time helps identify patterns and changes in its condition.

The system uses timestamps to organise sensor readings chronologically and provide additional information about streetlight performance.

The following time-series analysis features are included in the application:

- Fleet Fault Rate Over Time

The system analyses the fault rate across the monitored streetlight fleet over time. This provides an overview of how predicted faults change during the monitoring period.

The analysis can help identify periods where the number or proportion of predicted faults increases, indicating that further investigation may be necessary.

-Hourly Voltage and Current Analysis

The application calculates and displays average voltage and current measurements according to the hour of the day.

This analysis helps examine how electrical measurements vary throughout the day and provides an opportunity to identify unusual changes in sensor behaviour.

-Streetlight Asset Risk Ranking

Smart Light AI includes an asset risk ranking feature that identifies streetlights that may require additional attention.

The ranking uses monitoring information to highlight assets that appear more at risk. This helps support maintenance prioritisation by making potentially problematic streetlights easier to identify.

-Time-Series Visualisation

The dashboard presents time-series information through graphical representations, including fleet fault-rate trends, hourly electrical measurements and asset risk information.

These visualisations make it easier for users to understand changes in the monitored data without having to examine every sensor record individually.

- Relevance to the Proposed Solution

Time-series analysis supports the main objective of Smart Light AI by helping identify changes in streetlight conditions over time.

For example, repeated unusual voltage or current readings may indicate a possible electrical problem that requires investigation. By examining trends and comparing measurements, maintenance personnel can gain additional information when deciding which streetlights require attention.

The analysis therefore complements machine learning classification by providing a time-based view of streetlight performance.

5. Solution Techniques

Smart Light AI combines different techniques to identify streetlight faults, improve model performance, and support maintenance activities.

-Fault Classification

The machine learning model analyses sensor measurements and predicts one of the five streetlight condition categories.

This technique provides an automated method of identifying possible faults from electrical and environmental information.

- Data Augmentation

The project includes data augmentation experiments to investigate whether improving the representation of underrepresented fault categories can improve classification performance.

This is relevant because a dataset with uneven class distribution can cause a model to perform better on common categories while struggling to identify less frequent faults.

-Hyperparameter Tuning

Smart Light AI includes RandomizedSearchCV to investigate different Random Forest parameter combinations.

The tuning process uses macro F1-score as its scoring objective. This helps identify model configurations that perform more consistently across different fault categories.

-Threshold Optimisation

The system includes prediction threshold experiments designed to investigate whether the identification of Bulb Failure can be improved without causing an unacceptable reduction in overall accuracy.

This is relevant because failing to identify a faulty streetlight may delay maintenance.

-Cross-Validation

Stratified K-Fold cross-validation is used to evaluate candidate models across multiple data splits. This provides additional information about how consistently a model performs.

-Anomaly Detection

The application includes anomaly detection as an additional technique for identifying unusual sensor behaviour.

Anomaly detection complements the classification model by highlighting readings that may differ from expected patterns and require further investigation.

-Maintenance Prioritisation

The system uses monitoring results to support fault prioritisation and maintenance ticket generation. This helps transform the model's predictions into information that can support maintenance decisions.

- How these techniques improve the solution

The techniques are not included independently. They contribute to improving different aspects of the system.

Data augmentation investigates class representation, hyperparameter tuning investigates model configurations, threshold optimisation examines fault detection trade-offs, and cross-validation assesses consistency.

Together, these techniques provide a structured approach to developing and evaluating an AI model that is appropriate for streetlight fault detection.

6. Natural Language Processing, Speech Recognition or Speech Synthesis

Smart Light AI includes a Natural Language Processing chatbot called the Smart Light Assistant. The assistant allows users to interact with the monitoring system using ordinary text questions.

The chatbot uses an offline NLP approach that combines Term Frequency-Inverse Document Frequency (TF-IDF), Logistic Regression, and regular expression-based entity extraction.

- TF-IDF Text Processing

TF-IDF converts the words and phrases entered by the user into numerical features that can be processed by a machine learning classifier.

The chatbot uses both individual words and combinations of words through an n-gram configuration.

This allows the system to identify useful patterns in the wording of user questions.

- Intent Classification

A Logistic Regression classifier is trained to identify the purpose of a user's question.

The chatbot supports 13 predefined intent categories:

- Greeting.
- Goodbye.
- Help.
- Fleet status.
- Count faults.
- List faults.
- Asset status.
- Explain fault.
- Recommend action.
- Urgent faults.

- At-risk assets.
- Model performance.
- Sensor information.

The classifier uses the predicted intent to determine which type of response should be provided.

- Entity Extraction

Regular expressions are used to extract specific information from user questions, including streetlight asset numbers and fault types.

For example, when a user asks about a particular streetlight, the chatbot can extract its asset number and use it when preparing a response.

-Chatbot Evaluation

The chatbot includes a separate test dataset containing example questions that are not part of its training examples.

The evaluate function measures intent classification accuracy and identifies incorrectly classified examples. This provides a way to assess whether the assistant correctly understands different question formulations.

- Speech Synthesis

The application also includes browser-based speech synthesis. When the user enables the read-aloud option, the browser uses its Speech Synthesis functionality to read the latest chatbot response aloud.

This provides an alternative way for users to receive information from the assistant.

-Relevance to Smart Light AI

The NLP assistant is relevant because maintenance personnel may need to access monitoring information quickly without manually searching through records.

Users can ask questions about fault counts, individual assets, urgent faults and recommended maintenance actions.

The chatbot therefore makes the monitoring system easier to interact with and provides a practical communication interface for the proposed solution.

The current implementation supports text-based NLP and speech synthesis. It does not currently implement speech recognition for converting spoken questions into text.

7. Deep Learning

Smart Light AI incorporates a neural network candidate through the Multi-Layer Perceptron (MLP) implementation referenced in the model development module. A Multi-Layer Perceptron is an artificial neural network that processes input features through interconnected layers of artificial neurons. In the proposed solution, the neural network is intended to learn relationships between streetlight sensor measurements and their corresponding fault categories. The sensor measurements provide the input to the network, while the fault categories represent the classification targets. The MLP candidate is included in the model comparison process alongside traditional machine learning algorithms. This allows the project to investigate whether a neural network approach can provide useful classification performance for streetlight fault detection. The neural network approach is relevant because electrical sensor measurements may contain complex relationships that are useful for identifying different fault conditions. Its performance can be examined using the same classification evaluation techniques applied to

the other candidate models, including accuracy, precision, recall, F1-score and confusion matrix analysis.

The implementation details, including the network architecture, hidden layers, activation functions and training configuration, are provided by the project's deep learning module.

8. Other Features: Chatbot/Softbot

Smart Light AI includes an interactive chatbot, monitoring dashboard, and maintenance management features to make the AI solution practical and accessible.

The additional features are designed to support the main fault detection process rather than operate as unrelated functions.

-Smart Light Assistant

The chatbot provides an interface through which users can ask questions about the monitored streetlight fleet.

It can provide information about fault counts, asset status, urgent faults, sensor information, model performance, and recommended maintenance actions.

- Interactive Monitoring Dashboard

The Streamlit dashboard provides a central monitoring interface where users can examine streetlight readings, predicted fault categories and monitoring results.

It also provides access to model evaluation, time-series analysis and the chatbot.

-Simulated Live Monitoring

The system includes a simulated live monitoring mode that demonstrates how incoming sensor readings could be processed by the AI model.

This allows the project to demonstrate the monitoring process without requiring physical IoT devices or a connection to actual streetlight infrastructure.

- Maintenance Ticket Generation

When possible, faults are identified, and the application supports the creation of maintenance tickets.

These tickets provide a structured way of recording identified problems and organising maintenance activities.

- Simulated Municipal Maintenance Portal

The application includes a simulated municipal portal where maintenance tickets can be viewed and their statuses managed.

This demonstrates how AI-generated fault information could support a municipal maintenance workflow.

- Notification History

Smart Light AI includes a notification history section where prepared maintenance notifications can be stored and reviewed.

The feature demonstrates how maintenance information could be organised for communication. The current version prepares and stores notifications locally; it does not send actual emails to a municipality.

-Relevance of the Additional Features

The additional features improve the practical usefulness of the solution.

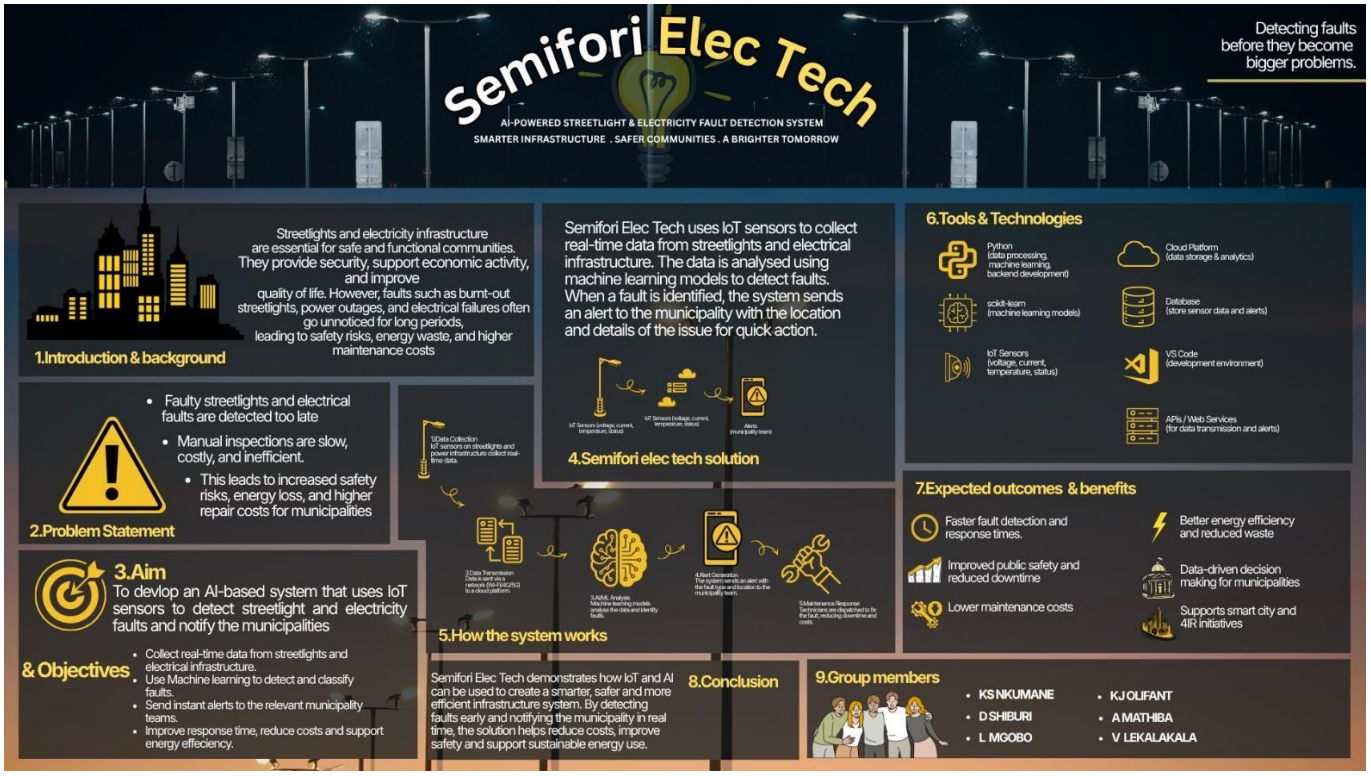
The monitoring dashboard presents the AI predictions, the chatbot makes the information accessible through natural language, and the maintenance features demonstrate how identified faults could be managed.

These features connect the technical AI components to the practical needs of streetlight maintenance.

The MLP neural network candidate demonstrates the inclusion of a neural network approach, and the additional dashboard and maintenance features show how AI predictions can be translated into useful operational information.

The overall solution demonstrates the relationship between the theoretical AI concepts and their practical application to streetlight fault detection and maintenance support.

Project poster



DECLARATION OF ORIGINALITY

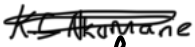
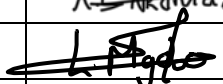
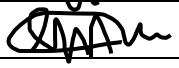
We hereby declare that the work submitted for the Semifori Elec Tech – Smart Light AI Project is our own original work.

We confirm that all information, ideas, research materials and other sources used in this project have been appropriately acknowledged and referenced where necessary.

We understand that plagiarism and presenting another person's work as our own constitute academic misconduct. We therefore accept responsibility for the originality and authenticity of the work submitted.

By signing below, all group members confirm that they have contributed to and agree with this declaration.

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SemiforiElecTech(Smartlight AI)

by Dzunani Shiburi

General metrics

34,819	4,384	399	17 min 32 sec	33 min 43 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 97% of all texts checked by Grammarly

Writing Issues

1		1
Issues left	Critical	Advanced

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

21%

unique words

Rare Words

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45%

rare words

Report: SemiforiElecTech(Smartlight AI)

Word Length

Measures average word length

6.6

characters per word

Sentence Length

Measures average sentence length

11

words per sentence